

Math 520, F07
Exam 2

Name: *Answers*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the quiz and exam.
- You must simplify your answer if possible.

Problem	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points) Find the general solution of the second-order homogeneous differential equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + \alpha y = 0$$

under the following cases:

(a) $\alpha = -5$.

Solution: characteristic equation:

$$r^2 + 4r - 5 = 0$$

$$(r+5)(r-1) = 0$$

$$r_1 = -5, r_2 = 1$$

$$\Rightarrow y_1(t) = e^{-5t}, y_2(t) = e^t$$

$$\begin{aligned}\Rightarrow y(t) &= c_1 y_1(t) + c_2 y_2(t) \\ &= c_1 e^{-5t} + c_2 e^t\end{aligned}$$

(b) $\alpha = 4$.

Solution: characteristic equation:

$$r^2 + 4r + 4 = 0$$

$$(r+2)^2 = 0$$

$$r_1 = r_2 = -2$$

$$\Rightarrow y_1(t) = e^{-2t}, y_2(t) = t e^{-2t}$$

$$\begin{aligned}\Rightarrow y(t) &= c_1 y_1(t) + c_2 y_2(t) \\ &= c_1 e^{-2t} + c_2 t e^{-2t}\end{aligned}$$

Problem 2. (20 points) Given a second-order nonhomogeneous differential equation

$$y'' + 4y = \csc 2t, \quad 0 < t < \pi/2.$$

(a) Find two linearly independent solutions $y_1(t)$ and $y_2(t)$ for the homogeneous equation $y'' + 4y = 0$ and compute the Wronskian $W[y_1, y_2](t)$.

Solution: characteristic equation:

$$r^2 + 4 = 0 \Rightarrow r^2 = -4$$

$$r_1 = 2i, \quad r_2 = -2i$$

$$\Rightarrow y_1 = \cos 2t, \quad y_2 = \sin 2t$$

$$W[y_1, y_2](t) = y_1(t) y_2'(t) - y_1'(t) y_2(t)$$

$$= \cos 2t (2 \sin 2t) - (-2 \sin 2t) \cos 2t = 2$$

(b) Find a particular solution $\psi(t)$ for the nonhomogeneous equation based on the results obtained from (a) using the **method of variation of parameters**.

Solution: $\chi(t) = u_1(t) y_1(t) + u_2(t) y_2(t)$

$$u_1(t) = \int \frac{-y_2(t) y_1'(t)}{W[y_1, y_2](t)} dt = \int \frac{-\sin 2t \cdot (-2 \cos 2t)}{2} dt$$

$$= \int -\frac{1}{2} dt = -\frac{t}{2}$$

$$u_2(t) = \int \frac{y_1(t) y_2'(t)}{W[y_1, y_2](t)} dt = \int \frac{\cos 2t \cdot 2 \sin 2t}{2} dt$$

$$= \int \frac{1}{2} \frac{\cos 2t}{\sin 2t} dt = \frac{1}{4} \int \frac{1}{\sin 2t} d(\sin 2t)$$

$$= \frac{1}{4} \ln |\sin 2t|$$

$$= \frac{1}{4} \ln(\sin 2t) \quad \text{Since } 0 < t < \frac{\pi}{2}.$$

$$\Rightarrow \chi(t) = -\frac{t}{2} \cos 2t + \frac{1}{4} \sin 2t \ln(\sin 2t)$$

(c) Write down the general solution of the nonhomogeneous equation based on the results obtained from (a) and (b).

Solution: $y(t) = C_1 y_1(t) + C_2 y_2(t) + \chi(t)$

$$= C_1 \cos 2t + C_2 \sin 2t - \frac{t}{2} \cos 2t + \frac{1}{4} \sin 2t \ln(\sin 2t)$$

Problem 3. (20 points) Given the following second-order nonhomogeneous differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = (1+2t)e^t.$$

Use **method of judicious guessing** to look for a particular solution $\psi(t)$ for this equation.

(a) Determine the general format of $\psi(t)$ for this equation.

Solution: Characteristic equation:

$$r^2 + 2r - 3 = 0$$

$$(r+3)(r-1) = 0$$

$$r_1 = -3, r_2 = 1$$

$$\Rightarrow \chi(t) = e^t t(A_0 + A_1 t)$$

(b) Find $\psi(t)$.

Solution: $\chi'(t) = e^t(A_0 + 2A_1 t) + e^t(A_0 t + A_1 t^2)$

$$= e^t[A_0 + (A_0 + 2A_1)t + A_1 t^2]$$

$$\chi''(t) = e^t[A_0 + (A_0 + 2A_1)t + A_1 t^2] + e^t[A_0 + 2A_1 + 2A_1 t]$$

$$= e^t[(2A_0 + 2A_1) + (A_0 + 4A_1)t + A_1 t^2]$$

$$\Rightarrow L[\chi] = \chi'' + 2\chi' - 3\chi$$

$$= e^t[4A_0 + 2A_1 + 8A_1 t]$$

$$= e^t(1 + 2t)$$

$$\Rightarrow \begin{cases} 4A_0 + 2A_1 = 1 \\ 8A_1 = 2 \end{cases} \Rightarrow A_1 = \frac{1}{4}, A_0 = \frac{1-2A_1}{4} = \frac{1}{8}$$

$$\Rightarrow \chi(t) = e^t t \left(\frac{1}{8} + \frac{t}{4} \right)$$

$$= e^t \left(\frac{t}{8} + \frac{t^2}{4} \right)$$

Problem 4. (20 points) Solve the initial-value problem

$$\frac{dx}{dt} = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 2 & -3 \\ 0 & -2 & 1 \end{pmatrix} x, \quad x(0) = \begin{pmatrix} 7 \\ 5 \\ 0 \end{pmatrix}.$$

Solution: $A = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 2 & -3 \\ 0 & -2 & 1 \end{pmatrix}$

$$P(\lambda) = \det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 0 & -3 \\ 0 & 2-\lambda & -3 \\ 0 & -2 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)[(2-\lambda)(1-\lambda) - 6] = (1-\lambda)(\lambda^2 - 3\lambda - 4)$$

$$= (1-\lambda)(\lambda-4)(\lambda+1) = 0$$

$$\Rightarrow \lambda_1 = 1, \lambda_2 = 4, \lambda_3 = -1.$$

(i) $\lambda_1 = 1$:

$$(A - I)V = 0 \Rightarrow \begin{pmatrix} 0 & 0 & -3 \\ 0 & 1 & -3 \\ 0 & -2 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -3v_3 = 0 \\ v_2 - 3v_3 = 0 \\ -2v_2 = 0 \end{cases} \Rightarrow v_2 = v_3 = 0$$

$$\Rightarrow V = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ 0 \\ 0 \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = C \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow x^1(t) = e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

(ii) $\lambda_2 = 4$:

$$(A - 4I)V = 0 \Rightarrow \begin{pmatrix} -3 & 0 & -3 \\ 0 & -2 & -3 \\ 0 & -2 & -3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow -3v_1 - 3v_3 = 0 \Rightarrow v_1 = -v_3$$

$$-2v_2 - 3v_3 = 0 \Rightarrow v_2 = -\frac{3}{2}v_3$$

$$\Rightarrow V = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} -v_3 \\ -\frac{3}{2}v_3 \\ v_3 \end{pmatrix} = v_3 \begin{pmatrix} -1 \\ -\frac{3}{2} \\ 1 \end{pmatrix} = C \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix}$$

$$\Rightarrow x^2(t) = e^{4t} \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix}$$

(iii) $\lambda_3 = -1$:

$$(A + I)V = 0 \Rightarrow \begin{pmatrix} 2 & 0 & -3 \\ 0 & 3 & -3 \\ 0 & -2 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2v_1 - 3v_3 = 0 \\ 3v_2 - 3v_3 = 0 \\ -2v_3 + 2v_3 = 0 \end{cases} \Rightarrow v_1 = \frac{3}{2}v_3, v_2 = v_3$$

$$\Rightarrow V = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} \frac{3}{2}v_3 \\ v_3 \\ v_3 \end{pmatrix} = v_3 \begin{pmatrix} \frac{3}{2} \\ 1 \\ 1 \end{pmatrix} = C \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

$$\Rightarrow x^3(t) = e^{-t} \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

$$\Rightarrow x(t) = C_1 x^1(t) + C_2 x^2(t) + C_3 x^3(t)$$

$$= C_1 e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + C_2 e^{4t} \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix} + C_3 e^{-t} \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

$$x(0) = \begin{pmatrix} 7 \\ 5 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} C_1 - 2C_2 + 3C_3 = 7 \\ -3C_2 + 2C_3 = 5 \\ 2C_2 + 2C_3 = 0 \end{cases}$$

$$\Rightarrow C_1 = 2, C_2 = -1, C_3 = 1$$

$$\Rightarrow x(t) = 2e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - e^{4t} \begin{pmatrix} -2 \\ -3 \\ 2 \end{pmatrix} + e^{-t} \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

Problem 5. (20 points) Given a matrix $A = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}$.

(a) Find two linearly independent solutions $\mathbf{x}^1(t)$ and $\mathbf{x}^2(t)$ of the system of differential equations $\dot{\mathbf{x}} = A\mathbf{x}$.

Solution:
$$\begin{aligned} p(\lambda) &= \det(A - \lambda I) \\ &= \det \begin{pmatrix} 3-\lambda & -2 \\ 4 & -1-\lambda \end{pmatrix} \\ &= (3-\lambda)(-1-\lambda) + 8 \\ &= \lambda^2 - 2\lambda + 5 \end{aligned}$$

$$\Rightarrow \lambda_1 = 1+2i, \lambda_2 = 1-2i$$

i) $\lambda_1 = 1+2i$

$$(A - (1+2i)I)V = 0 \Rightarrow \begin{pmatrix} 2-2i & -2 \\ 4 & -2-2i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow (2-2i)v_1 - 2v_2 = 0 \Rightarrow v_2 = (1-i)v_1$$

$$4v_1 - (2+2i)v_2 = 0$$

$$\Rightarrow V = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 \\ (1-i)v_1 \end{pmatrix} = v_1 \begin{pmatrix} 1 \\ 1-i \end{pmatrix} = c \begin{pmatrix} 1 \\ 1-i \end{pmatrix}$$

$$\Rightarrow \tilde{\mathbf{x}}(t) = e^{(1+2i)t} \begin{pmatrix} 1 \\ 1-i \end{pmatrix}$$

$$= e^t (\cos 2t + i \sin 2t) \begin{pmatrix} 1 \\ 1-i \end{pmatrix}$$

$$= e^t \begin{pmatrix} \cos 2t + i \sin 2t \\ \cos 2t + \sin 2t + i(\sin 2t - \cos 2t) \end{pmatrix}$$

$$\Rightarrow \mathbf{x}^1(t) = \operatorname{Re}(\tilde{\mathbf{x}}(t))$$

$$= e^t \begin{pmatrix} \cos 2t \\ \cos 2t + \sin 2t \end{pmatrix}$$

$$\mathbf{x}^2(t) = \operatorname{Im}(\tilde{\mathbf{x}}(t))$$

$$= e^t \begin{pmatrix} \sin 2t \\ \sin 2t - \cos 2t \end{pmatrix}$$

Extra credits (5 points)

(b) Then $X(t) = [x^1(t) \ x^2(t)]$ is a fundamental matrix solution of $\dot{x} = Ax$. Compute e^{At} based on this fact.

$$\text{Solution: } X(t) = \begin{pmatrix} e^{t \cos 2t} & e^{t \sin 2t} \\ e^{t(\cos 2t + \sin 2t)} & e^{t(\sin 2t - \cos 2t)} \end{pmatrix}$$

$$\Rightarrow X(0) = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$$

$$[X(0)]^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$$

$$\begin{aligned} \Rightarrow e^{At} &= X(t) [X(0)]^{-1} \\ &= \begin{bmatrix} e^{t(\cos 2t + \sin 2t)} & -e^{t \sin 2t} \\ 2e^{t \sin 2t} & e^{t(\cos 2t - \sin 2t)} \end{bmatrix} \end{aligned}$$